Exact Neural Spike Train Inference via Mixed Integer Quadratic Programming

- presenter: Jongmin Mun
- Based on: Lee, J., Gómez, A., and Atamtrk, A. (2024). Convexification of Multi-period Quadratic Programs with Indicators.

Spike train inference and calcium imaging

Optimization problem

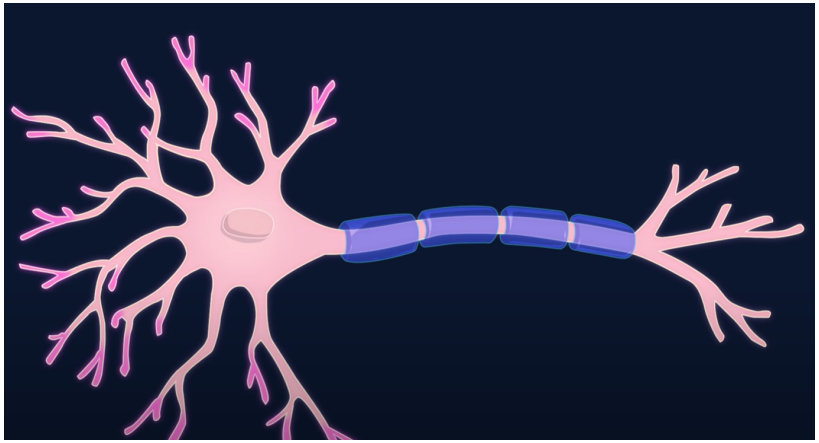
How to solve the problem

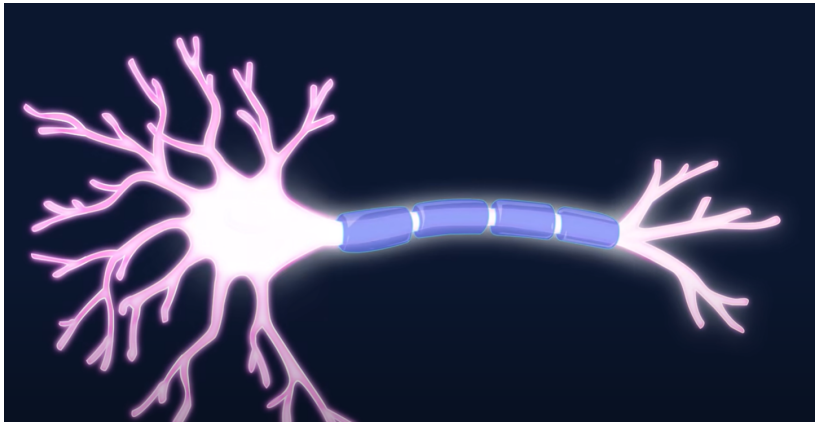
How to set the hyperparameters

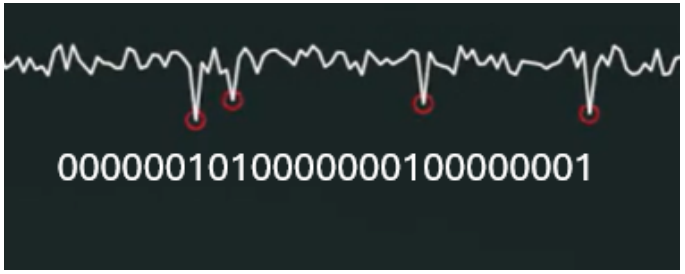
Benchmarking

Final Report

Spike train inference: from brain activity to binary code

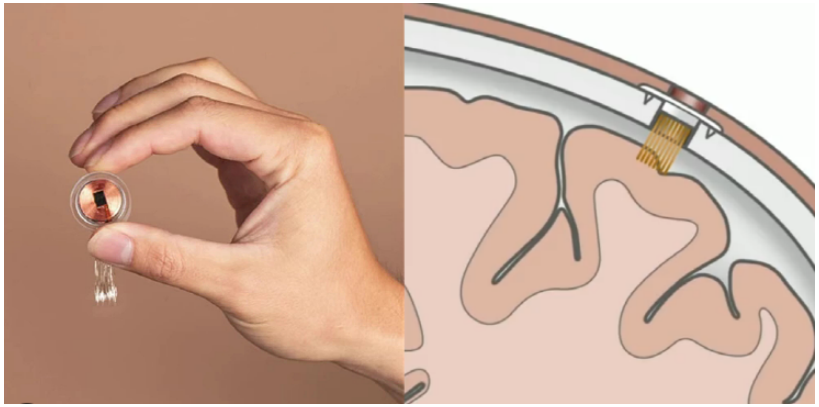






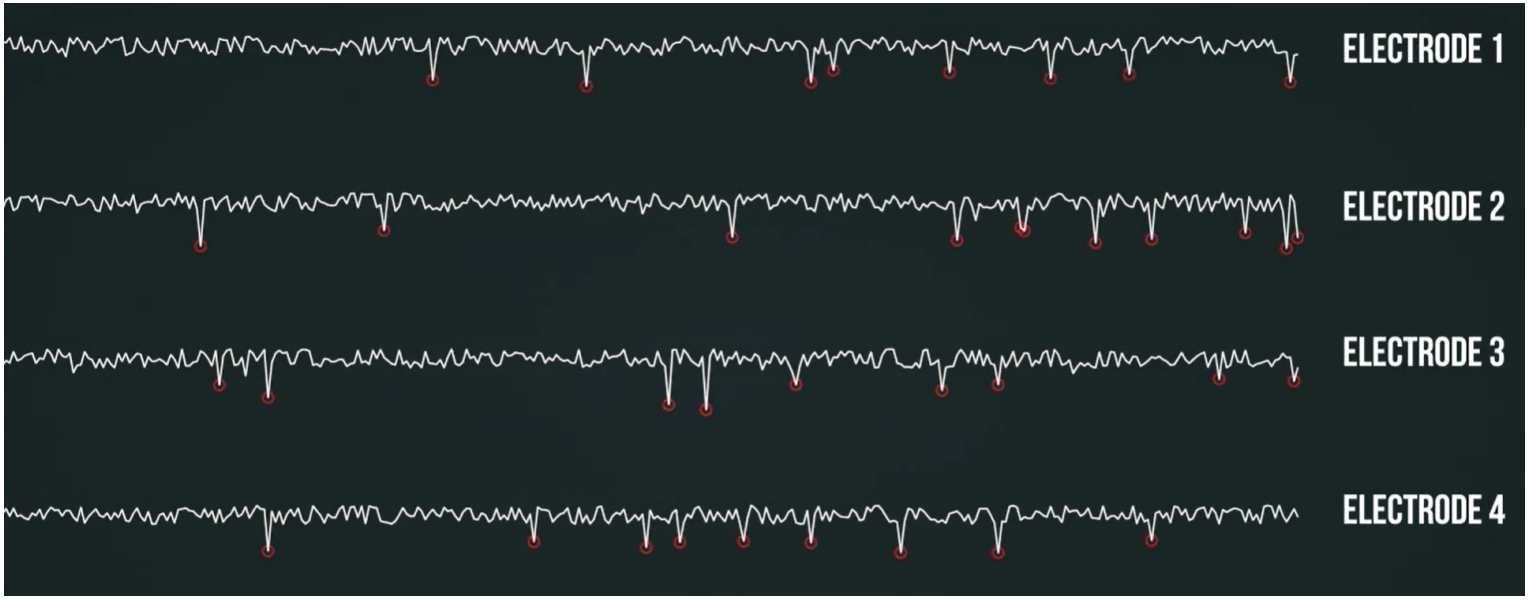
- Brain operates through electrical events known as *action potentials*
- Information is primarily encoded as sequences of these action potentials: binary patterns (01010...) known as *spike trains*
- *Spike train inference*: recovering these spike sequences from recorded neural electrical signals

Connection to ML



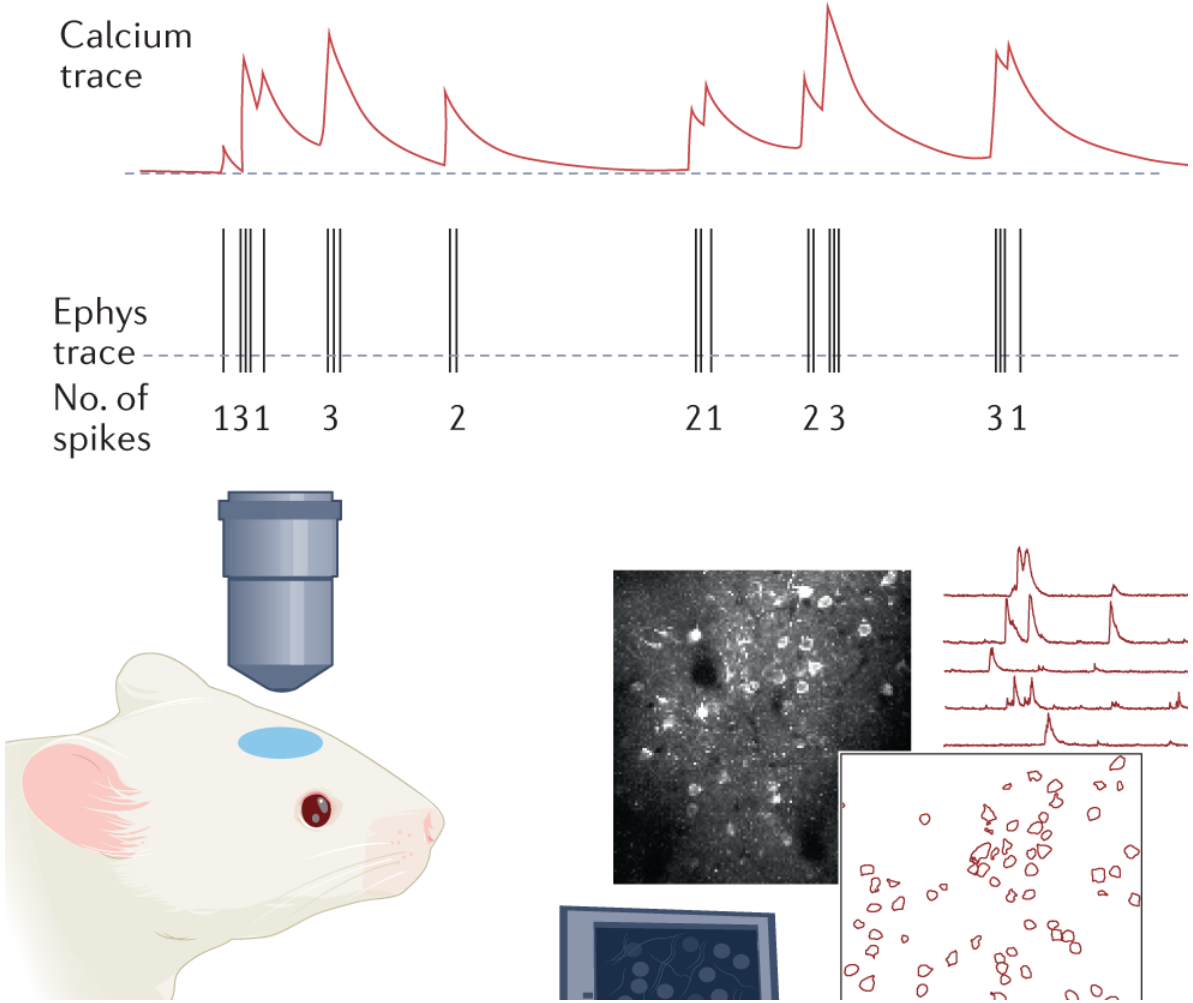
- Spike train inference empowers machine learning tasks such as both *brain decoding* (inferring thoughts from neural activity) and *brain encoding* (translating intent into neural signals for action)

Electrodes



- Spike train inference is relatively straightforward when high-quality electrical signals are directly recorded (e.g., via simple thresholding methods)
- However, such recordings are often invasive and do not scale well to monitoring large populations of neurons simultaneously

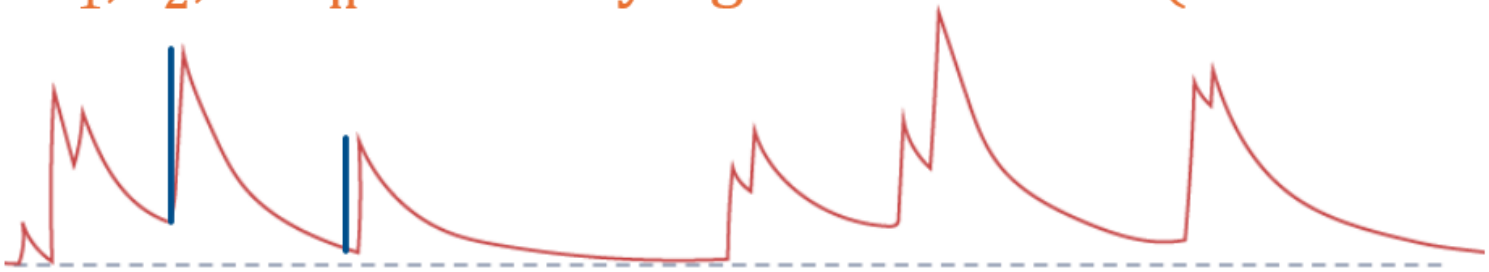
Calcium imaging: slow but scalable



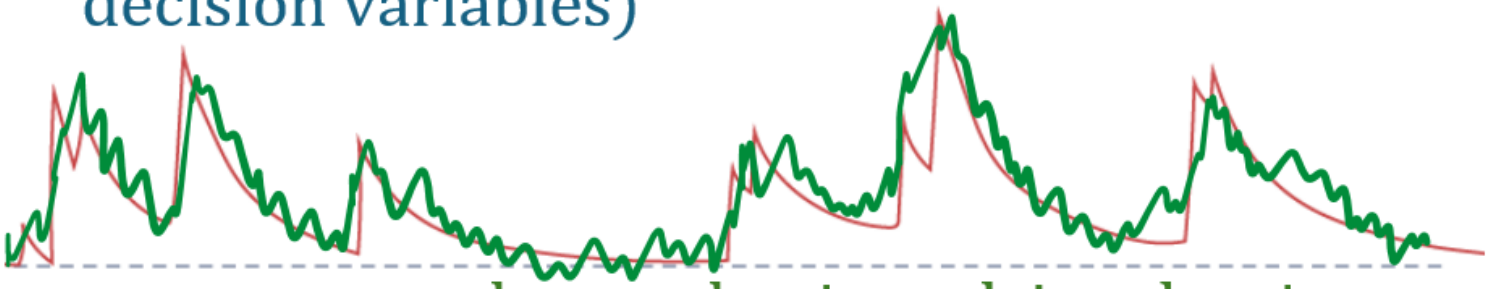
- Calcium imaging records thousands of neurons simultaneously, but its slow dynamics cause signals to overlap
- new spikes can occur before previous spikes have fully decayed

Mixed Integer Programming Formulation

$s_1, s_2, \ldots s_n$: underlying calcium level (decision variables)



$x_1, x_2, \ldots x_n$: effect of action potential (0 if no spike; decision variables)



$r_1, r_2, \ldots r_n$: observed noisy calcium level

To filter out the noise, we solve least squares:

$$\min_{\mathbf{s}} \frac{1}{2} \sum_{i=1}^{n+1} (s_i - r_i)^2$$

Mixed Integer Programming Formulation

- AR(1) model:

$$s_{i+1} = \alpha s_i + x_i$$

  - Calcium level always decays by $\alpha \in (0, 1)$
  - If $x_i > 0$, there is an effect of an action potential
- Sparse spikes: add an on-off switch $z_1, \ldots, z_n$ and link it to x_i:

$$\begin{aligned}
\min_{\mathbf{s},\mathbf{x},\mathbf{z}} \quad & \frac{1}{2}\sum_{i=1}^{n+1}(s_i - r_i)^2 + \lambda \sum_{i=1}^{n} z_i \\
\text{s.t.} \quad & x_i(1 - z_i) = 0, \quad z_i \in \{0, 1\}, \quad i = 1, \ldots, n \\
& x_i = s_{i+1} - \alpha s_i, \quad i = 1, \ldots, n \\
& s_1 = 0
\end{aligned}$$

How to solve this problem

- We employee the convexification framework of Lee, J., Gómez, A., and Atamtrk, A. (2024). Convexification of Multi-period Quadratic Programs
- This shows that for MIQP with special structure, convex relaxation yields MIP solution
- This is a general framework for an abstract problem; but can be applied to spike train inference

Convexification

The linear dynamic $x_i = s_{i+1} - \alpha s_i$ allows us to project out s_i:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z}} \quad & \mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{a}^\top \mathbf{x} + \lambda \sum_{i=1}^{n} z_i \\ \text{s.t.} \quad & \mathbf{x} \circ (\mathbf{e} - \mathbf{z}) = 0 \quad \mathbf{z} \in \{0,1\}^n, \end{aligned}$$

where $\mathbf{Q}$ is symmetric and *decomposable:* $q_{ij} = u_i v_j$ for $i \leq j$,

$$u_i := \alpha^{n-i}, \quad v_i := \frac{\alpha^{2(n-i+1)} - 1}{2\alpha^{n-i}(\alpha^2 - 1)},$$

$$a_i := -\sum_{t=i+1}^{n+1} \alpha^{t-(i+1)} r_t.$$

Convexification

epigraph formulation:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z},\tau} \quad & \tau + \mathbf{a}^\top \mathbf{x} + \lambda \sum_{i=1}^{n} z_i \\ \text{s.t.} \quad & (\mathbf{x}, \mathbf{z}, \tau) \in X_Q, \end{aligned}$$

where

$$\begin{aligned} X_Q := \{ & (\mathbf{x}, \mathbf{z}, \tau) \in \mathbb{R}^n \times \{0,1\}^n \times \mathbb{R}_+ : \\ & \tau \geq \mathbf{x}^\top \mathbf{Q} \mathbf{x},\ x_i(1 - z_i) = 0, i \in [n] \} \end{aligned}$$

Convexification

$$\text{cl conv}(X_Q) = C \cap P,$$

where

- C do not restrict $\mathbf{z}$:

$$C := \{(\mathbf{x}, \mathbf{z}, \tau) \in \mathbb{R}^{2n+1} : \exists \mathbf{W} \in \mathbb{R}^{n \times n}, \mathbf{w} \in \mathbb{R}_+^{(n+1)(n+2)/2} \text{ such that } \begin{bmatrix} \mathbf{W} & \mathbf{x} \\ \mathbf{x}^\top & \tau \end{bmatrix} \succeq 0\},$$

- P is a polytope that restricts $\mathbf{z}$ whose corners are integral in $\mathbf{z}$.

Convexification

- Therefore, the solution of the convex relaxation

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z},\tau} \quad & \tau + \mathbf{a}^\top \mathbf{x} + \lambda \sum_{i=1}^{n} z_i \\ \text{s.t.} \quad & (\mathbf{x}, \mathbf{z}, \tau) \in \text{cl conv}(X_Q), \end{aligned}$$

  is feasible for the original problem.

- This relaxation can be reformulated into shortest path problem for DAG with arc cost which can be solved in $O(n^2)$:

$$\ell_{ij} = \begin{cases} 0, & i = 0,\ 1 \le j \le n+1 \\ \lambda - \dfrac{(1-\alpha^2)\left(a_i - \alpha^{j-i} a_j\right)^2}{2\left(1-\alpha^{2(j-i)}\right)}, & 1 \le i < j \le n \\ \lambda - \dfrac{(1-\alpha^2) a_i^2}{2\left(1-\alpha^{2(n-i+1)}\right)}, & 1 \le i \le n,\ j = n+1. \end{cases}$$

Hyperparameters

- Decay parameter of AR(1): α
- Observational noise: σ^2
- Sparsity parameter: λ

Practitioners set these based on statistical model and domain knowledge

The AR(1) paramter α

- $\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t$
- empirical autocovariance at lag τ is estimated as:

$$\hat{C}_r(\tau) = \frac{1}{N-\tau}\sum_{t=1}^{N-\tau}(r_t - \bar{r})(r_{t+\tau} - \bar{r}).$$

- Assuming the underlying spiking signal $\mathbf{x}$ originates from a homogeneous Poisson process, and AR(1), we have $C_r(\tau) = \alpha C_r(\tau - 1)$, which yields an estimator:

$$\hat{\alpha} = \frac{\hat{C}_r(2)}{\hat{C}_r(1)}.$$

Observational noise σ^2

- Calcium dynamics are smooth and slow, and therefore primarily concentrated in low-frequency components.
- Random jitter from the microscope sensor is distributed uniformly across all frequencies.
- Therefore, σ^2 is estimated by isolating the high-frequency components and computing their variance.

Sparsity parameter λ

- We select λ such that the residual sum of squares satisfies

$$\sum_{i=1}^{n}(s_i - r_i)^2 \approx n\sigma^2$$

- For a given λ, we solve the optimization problem and compute the resulting mean squared error (MSE).
- We then compare the MSE to $n\hat{\sigma}^2$ and use a bisection search over λ until the target value is reached.

Datasets and Benchmarking

- 7 real datasets covering multiple species (mouse, zebrafish), brain regions, and calcium indicators
- They are ground truth datasets; scientists gave shocks to the animal and recorded the timings
- Benchmarked against a dynamic programming method (*Jewell et al. 2020*).
- Both methods are exact solvers
- Evaluation metrics:
  - Precision: $\frac{TP}{TP+FP}$
  - Recall: $\frac{TP}{TP+FN}$
  - F1 score: harmonic mean of precision and recall

Precision

Dataset	Precision (spp)	Precision (dp)
OGB1-m-V1	0.3607	**0.7087**
OGB1-2-m-V1	0.0754	**0.5200**
Cal520-m-S1	**0.1349**	0.0353
Cal520-zf-pDp	0.3049	**0.4468**
GCaMP6f-zf-aDp	0.3670	**0.7056**
GCaMP6f-zf-dD	0.0337	**0.1385**
GCaMP6f-m-V1	**0.1698**	0.1208

Precision comparison (best per row in bold).

Recall

Dataset	Recall (spp)	Recall (dp)
OGB1-m-V1	**0.6364**	0.4826
OGB1-2-m-V1	**1.0000**	0.8387
Cal520-m-S1	**1.0000**	**1.0000**
Cal520-zf-pDp	**0.9804**	0.5122
GCaMP6f-zf-aDp	**0.9877**	0.5031
GCaMP6f-zf-dD	**1.0000**	0.8571
GCaMP6f-m-V1	0.9931	**1.0000**

Recall comparison (best per row in bold).

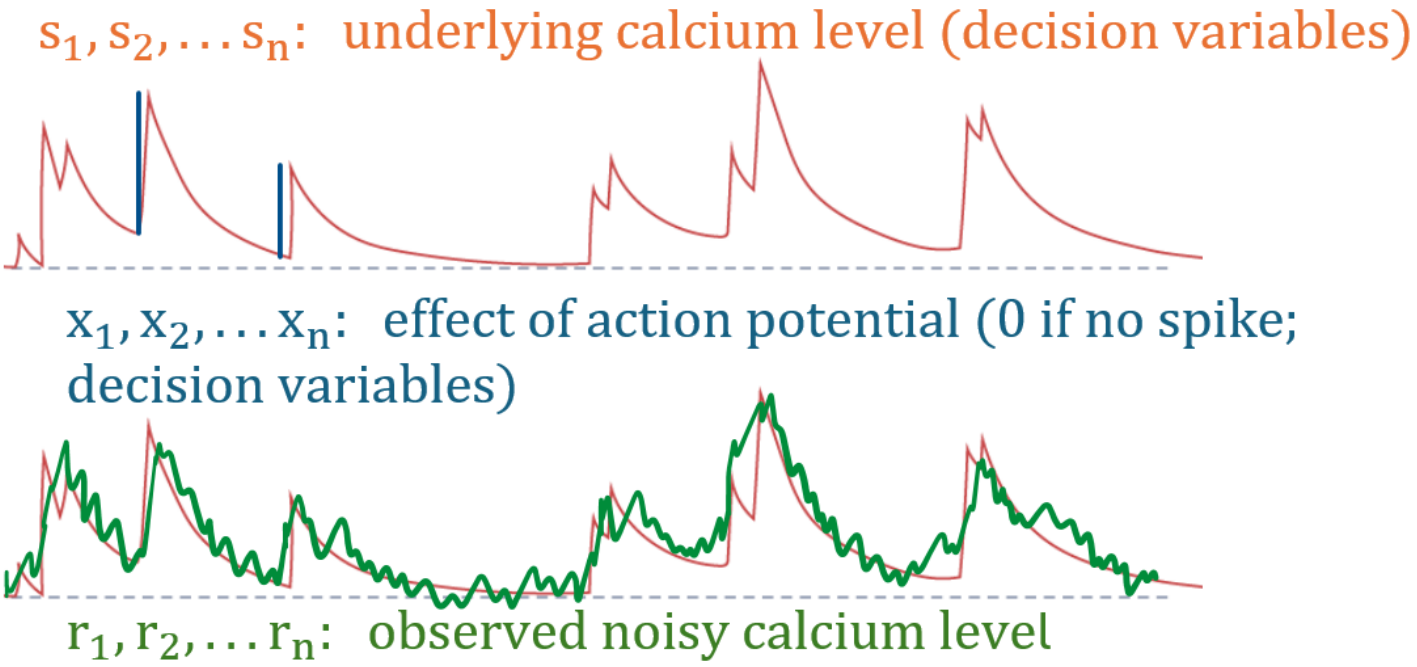
F1 Score

Dataset	F1 (spp)	F1 (dp)
OGB1-m-V1	0.4604	**0.5742**
OGB1-2-m-V1	0.1402	**0.6420**
Cal520-m-S1	**0.2378**	0.0682
Cal520-zf-pDp	0.4651	**0.4773**
GCaMP6f-zf-aDp	0.5351	**0.5874**
GCaMP6f-zf-dD	0.0652	**0.2384**
GCaMP6f-m-V1	**0.2901**	0.2155

F1 score comparison (best per row in bold).

- SPP wins on recall but loses on precision: it is overestimating
- Both methods are exact, thus the result is due to λ
- If we want to use SPP this in practice, we should find a better way to tune λ
- SPP (solved by Python igraph pacakge) is slower than Jewell's method written in C++, but still fast enough to solve 10000 variables in seconds

For the Final Report



- $x_1, \ldots, x_n$ are positive, so the constraint $x_i = s_{i+1} - \alpha s_i$ should be $x_i = s_{i+1} - \alpha s_i \geq 0$
- We can also add additional constraints e.g. capacity constraint
- Then SPP formulation does not work. Instead, we have SOCP formulation. We can also evaluate this on real dataset
- Might need to use Lagrangian Decomposition

Wrap up

- Neural spike train inference on calcium imaging data is mixed integer quadratic programming
- Convexification technique of Lee et al (2024), which was developed for general purpose, can successfully solve this problem